MILITARY SPECIFICATION

BURS, SURGICAL, TUNGSTEN CARBIDE, GENERAL REQUIREMENTS FOR

This specification has been prepared by the Defense Personnel Support Center, Defense Logistics Agency, based upon currently available technical information but it has not been approved for promulgation as a coordinated specification. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This document covers general requirements for tungsten carbide, surgical burs used in DoD Deployable Medical Systems (DMS) and is intended to be used in conjunction with the detailed specification sheets or procurement documents covering such additional requirements as are applicable to each individual bur.

1.2 Classification.

1.2.1 Head shapes and shank lengths. The surgical burs shall be of the following head shapes and shank lengths as specified in the applicable specification sheets (see 6.2):

Shape:

A. Round Cutting
B. Side Cutting, Straight
C. Side Cutting, Tapered
D. Oval Cutting
E. Barrel Cutting

Shank Length:

1. Short
2. Medium
3. Long
4. Extra Long

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Methods of Preservation.

FEDERAL

PPP-B-566	-	Box, Folding, Paperboard.
PPP-B-636	-	Box, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

FEDERAL

FED-STD-151	-	Metals: Test Methods.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. All burs shall conform to the requirements specified herein and the specification sheets applicable to the individual bur specified.

3.1.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

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3.2 Design and construction. The burs shall be surgical, tungsten carbide burs suitable for use in bone surgery when placed in the Pneumatic Bone Drill, National Stock Number (NSN) 6515-01-116-8197. The bur head shall have properly and uniformly sharpened cutting edges. The style, design, and dimensions of the burs shall be as specified in the detail specification sheets for the applicable NSN (see 6.2).

3.3 Runout. The total indicated runout of the burs shall not exceed .002 inch.

3.4 Finish. The bur head shall have a finely honed finish with sharpened blades. The neck and shank of the burs shall be smooth, and any safeline present on the shank shall have an etched finish.

3.5 Autoclaving. The burs shall show no corrosion when tested as specified in 4.5.3.

3.6 Workmanship. The burs shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2.1 Certificate of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.3 Examination.

4.3.1 Visual examination. Visual examination for defects shall be conducted at an inspection level of II with an AQL of 1.0.

4.3.2 Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used, if properly calibrated checking standards are used in conjunction with them. The inspection level shall be S-2 with an AQL of 2.5.

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4.3.3 Microscopic examination. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 30 power (30X) magnification, unless otherwise specified herein. The inspection level shall be S-1 with an AQL of 2.5.

4.4 Packaging inspection. The inspection of the unit package, packing, and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests.

4.5.1 Chemical composition. In the absence of certificates of quality for the raw material, the samples shall be analyzed in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be used for referee analysis. Tests shall be made on specimens representing each lot of carbide and the results averaged per lot.

4.5.2 Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator, sensitive to .0002 inch and equipped with a knife-edge foot, a shadow-graph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360° rotation of the bur shall be measured and recorded to the nearest .0005 inch. The difference between these two readings shall be taken as the total indicated runout. The inspection level shall be S-1 with an AQL of 1.5.

4.5.3 Autoclaving. The bur shall be autoclaved at $270^{\circ} \pm 5^{\circ}$ and 27 ± 1 pounds per square inch of pressure for 20 minutes. The inspection level shall be S-1 with an AQL of 1.0.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Unit of issue. Each (EA). One bur, as specified, constitutes one unit.

5.1.2 Level A.

5.1.2.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.2.2 Preservative. Preservative shall not be utilized.

5.1.2.3 Unit package. One bur of the shape and length specified, shall be packaged in a box of appropriate size so designed that each bur is contained in an individual cell. The box shall be constructed of an anticorrosive material that will not interact with the contents. Closure of the box shall be secured to prevent spilling of contents under normal handling.

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5.1.2.4 Intermediate package. Twelve unit packages shall be packed in snug-fitting box constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors are acceptable. Closure of the box shall be secured to prevent spilling of contents under normal handling.

5.1.3 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twenty-four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twenty-four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Marking. Each unit package, intermediate package, and shipping container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. Tungsten carbide, surgical burs covered by this specification are intended for use in medical war readiness assemblages and DoD Deployable Medical Systems.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition document should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of the applicable specification sheet.
c. NSN (as applicable).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Agent:
DLA-DM

Preparing activity:
DoD-MB

Project No. DEPMEDS-002

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-B-37959(DM) BURS, SURGICAL, TUNGSTEN CARBIDE, GENERAL REQUIREMENTS FOR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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